

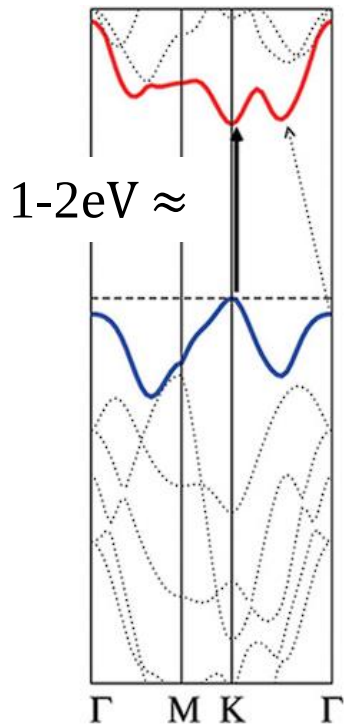
Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

Student : 劉人瑩

Supervisor : 鄭舜仁 教授

1. Introduction

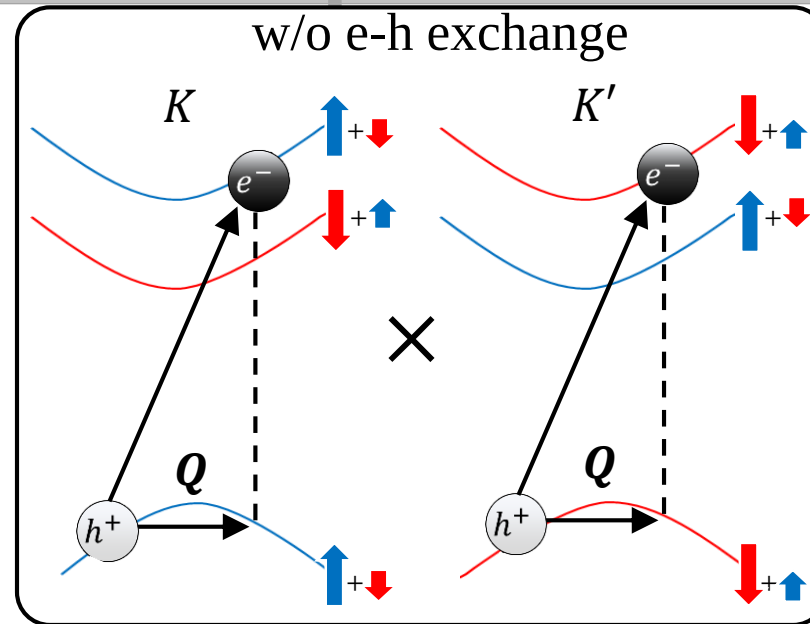
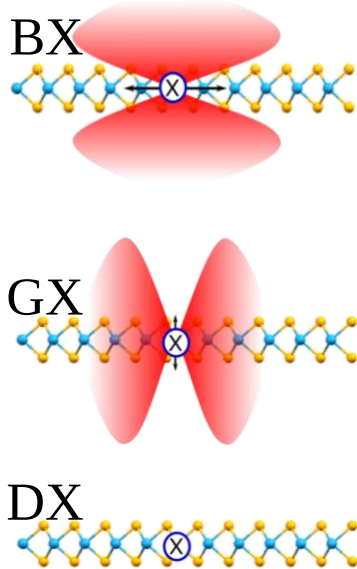
2D Transition metal dichalcogenide (TMD)



Direct band gap
&
Suitable energy gap

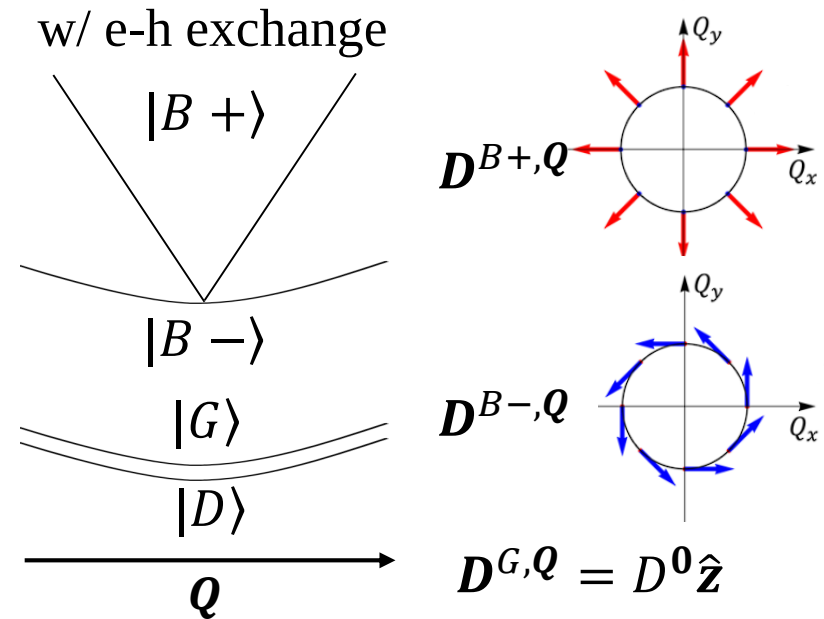
A. Splendiani *et al.*, *Nano Lett.* **10**, 1271 (2010)

Spin-orbit interaction in 2D TMD



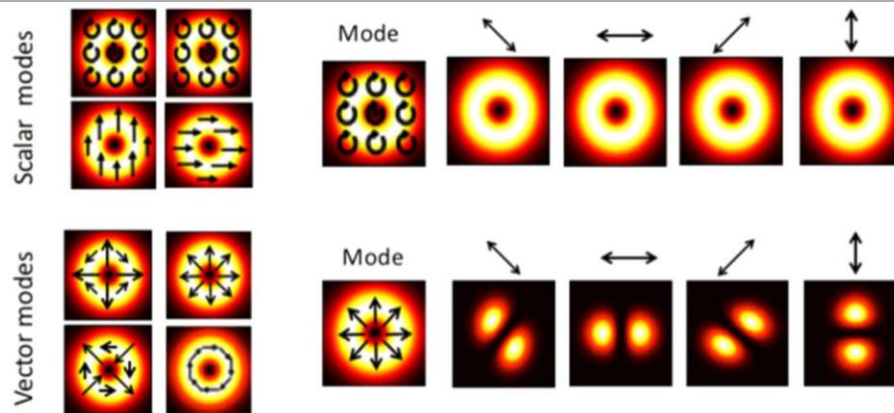
L. M. Schneider *et al.*, *Sci. Rep.* **10**, 8091 (2020)

Valley-mixing in 2D TMD



D. Y. Qiu *et al.*, *Phys. Rev. Lett.* **115**, 176801 (2015) / G.-H. Peng *et al.*, *Phys. Rev. B* **106**, 155304 (2022).

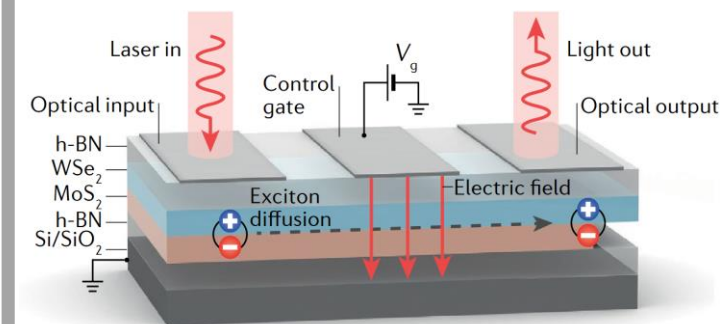
Manipulation in light – vortex beam



C. Rosales-Guzmán, B. Ndagano *et al.*, *J. Opt.* **20**, 123001 (2018)

Manipulation of
Polarization
&
Intensity

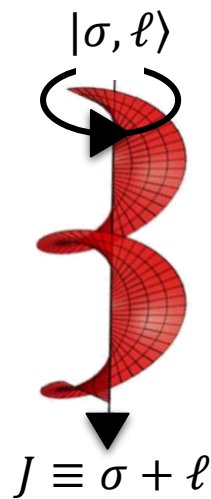
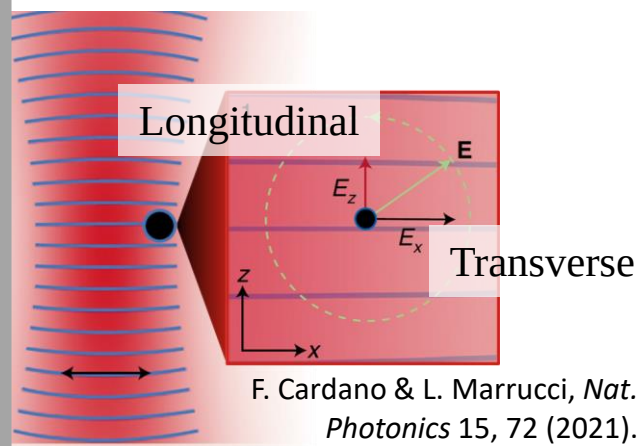
Device based on exciton diffusion



A. Ciarrocchi, F. Tagarelli *et al.*, *Nat. Rev. Mater.* **7**, 449 (2022)

1. Introduction

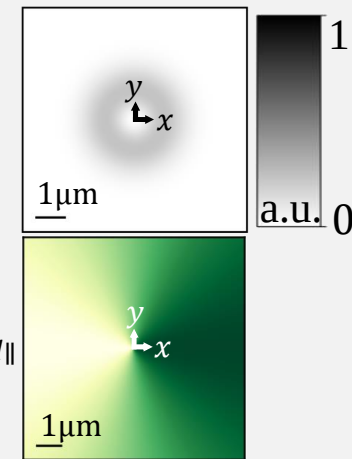
Spin-orbit interaction (SOI) in paraxial light



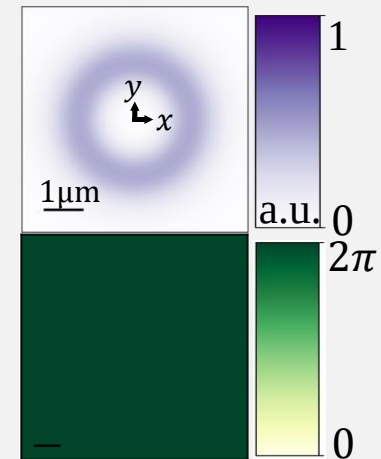
Intensity :

Phase :

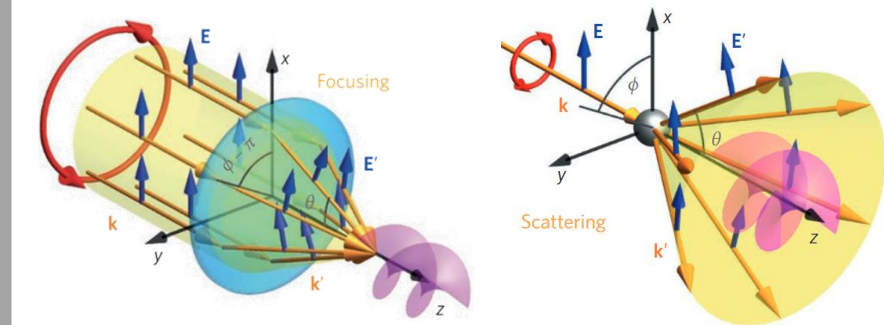
Transverse



Longitudinal



SOI in nonparaxial fields

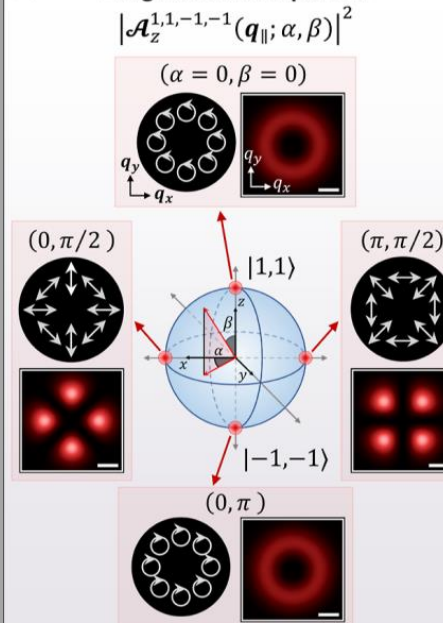


$$S = \sigma \cos \theta \frac{\mathbf{P}}{P}, \quad L = [\ell + \sigma (1 - \cos \theta)] \frac{\mathbf{P}}{P}$$

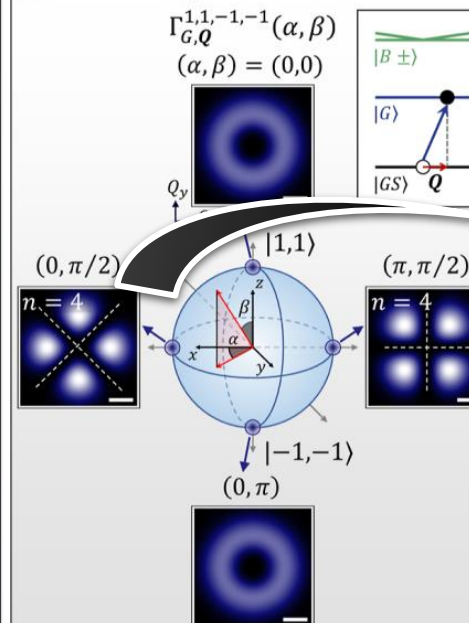
K. Y. Bliokh, F. J. Rodríguez-Fortuño *et al.*, *Nat. Photonics* 9, 796–808 (2015)

Selective excitation with SOI in light

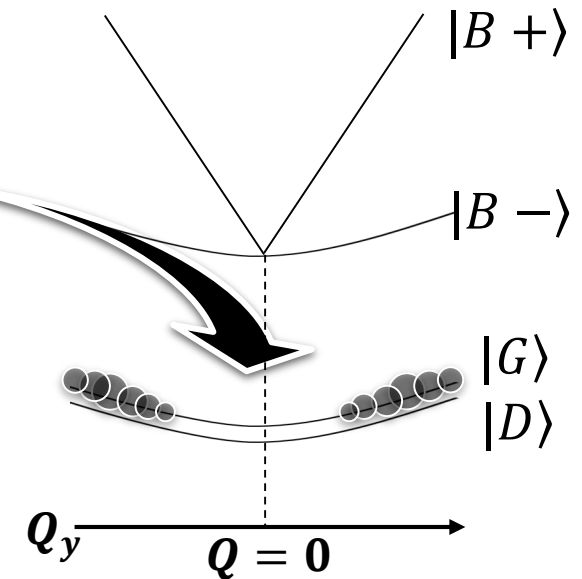
(b) Longitudinal Component



(b) Gray Exciton Band



O. J. G. Sanchez, G.-H. Peng *et al.*, *ACS Nano* 18, 11425 (2024).



1. Introduction

2. Vector vortex beam (VVB) : SOI beyond single beams

- Scalar vortex beams in angular spectrum & conditions
- VVB : even number of AM difference - symmetry
- VVB : odd number of AM difference - asymmetry

3. Optical transitions

- Directional momentum-selective excitation
- Selective excitation

Summary and outlook

1. Introduction

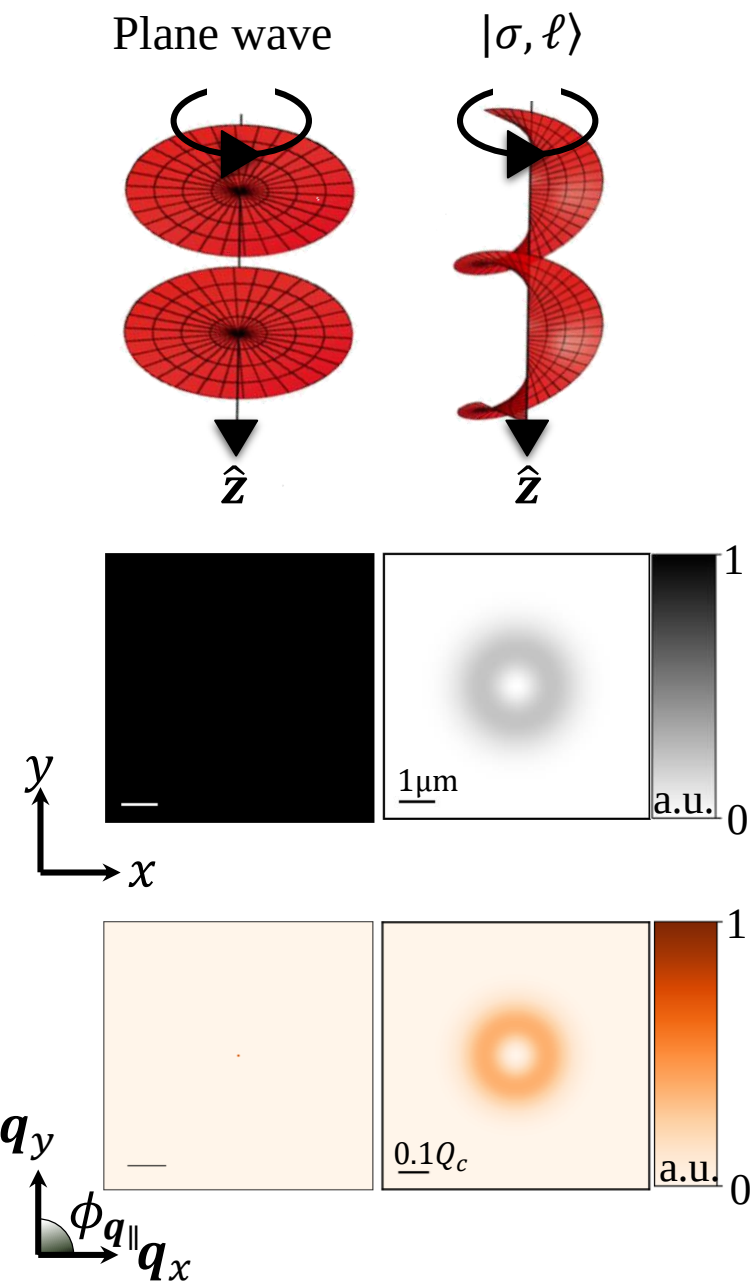
2. Vector vortex beam (VVB) : SOI beyond single beams

- Scalar vortex beams in angular spectrum & conditions
- VVB : even number of AM difference - symmetry
- VVB : odd number of AM difference - asymmetry

3. Optical transitions

Summary and outlook

2. Scalar vortex beams in angular spectrum



Vector potential form in Coulomb gauge :

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) = \tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel})\hat{\mathbf{e}}_{\sigma} + \tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel})\hat{\mathbf{z}}$$

$$\hat{\mathbf{e}}_{\sigma} \equiv (\hat{\mathbf{x}} + i\sigma\hat{\mathbf{y}})/\sqrt{2}, \sigma = \pm 1$$

Transverse :

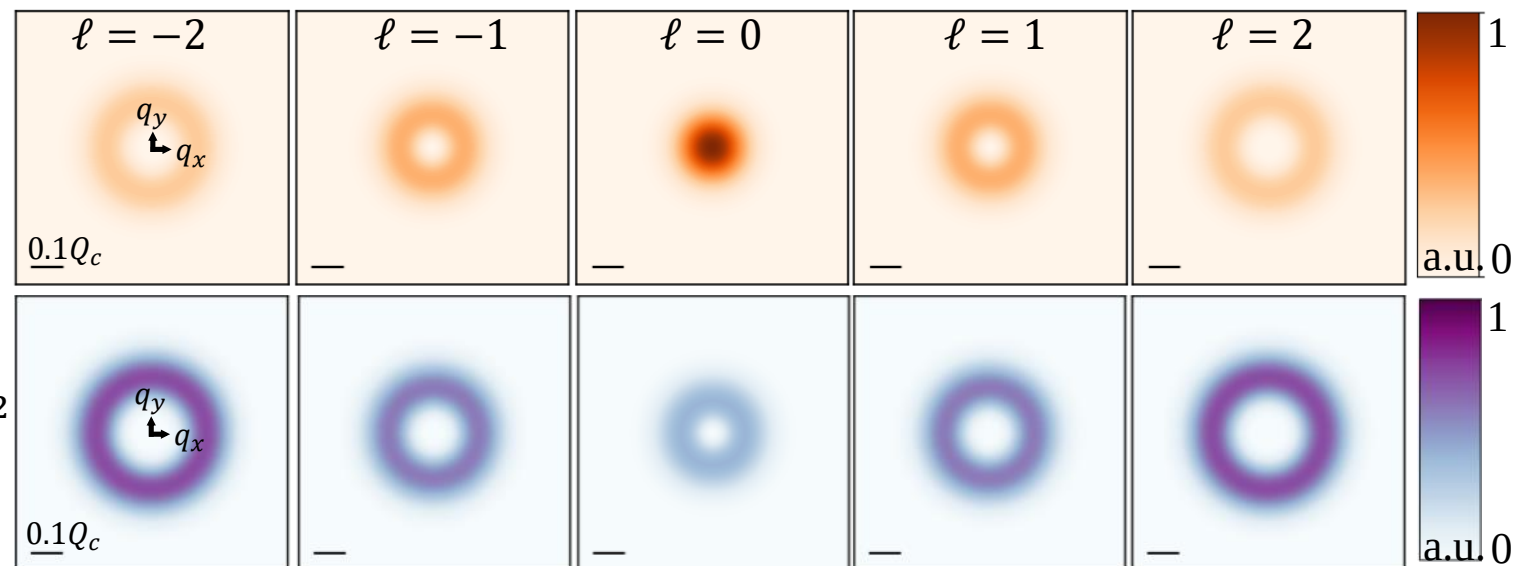
$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel})e^{i\ell\phi_{q_{\parallel}}}$$

Longitudinal :

$$\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel}) \approx \frac{-\sin\theta_q}{\sqrt{2}}\tilde{F}_{|\ell|}(\mathbf{q}_{\parallel})e^{i\ell\phi_{q_{\parallel}}}$$

$$|\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel})|^2$$

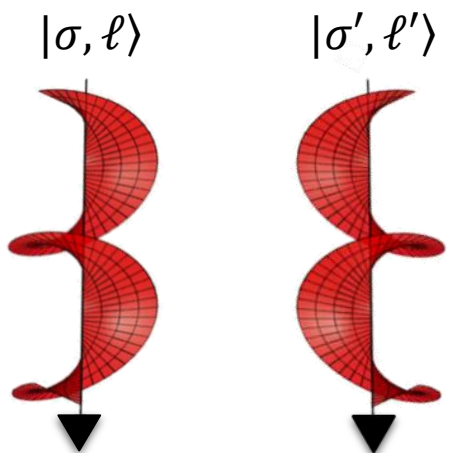
$$|\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel})|^2$$



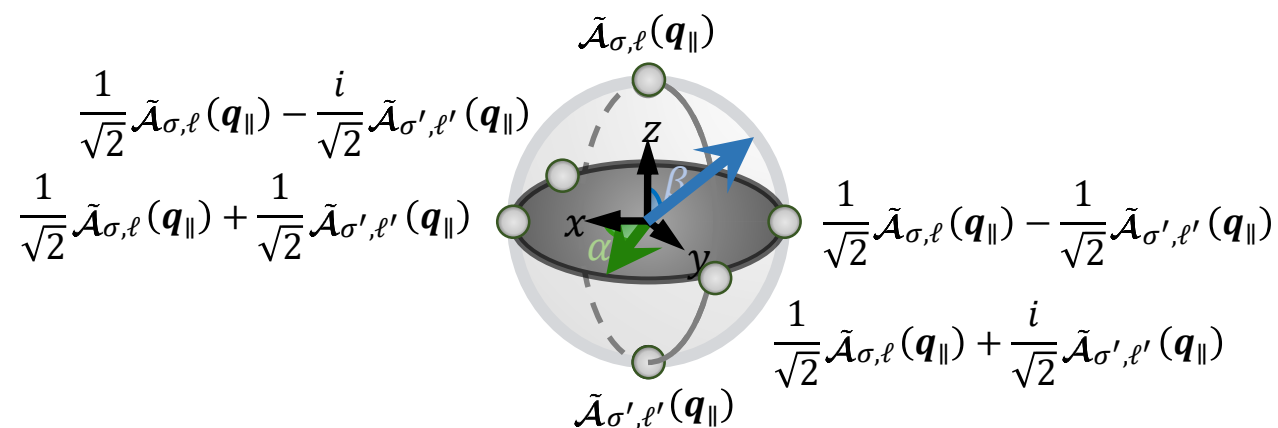
2. Vector vortex beams (VVB)

Vector vortex beams :

$$\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) = \cos\left(\frac{\beta}{2}\right) \tilde{\mathcal{A}}_{\sigma,\ell}(\mathbf{q}_{\parallel}) + e^{i\alpha} \sin\left(\frac{\beta}{2}\right) \tilde{\mathcal{A}}_{\sigma',\ell'}(\mathbf{q}_{\parallel})$$



Represent in Poincaré sphere



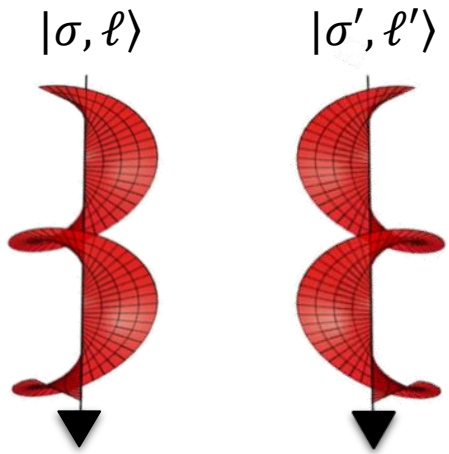
2. VVB : focus on $\beta = \pi/2$ and $\alpha = 0$

Vector vortex beams :

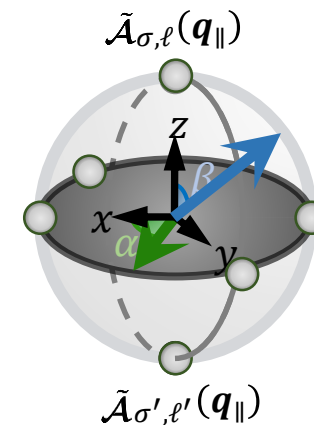
$$\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha = 0, \beta = \frac{\pi}{2}) = \frac{1}{\sqrt{2}} \tilde{\mathcal{A}}_{\sigma',\ell}(\mathbf{q}_{\parallel}) + \frac{1}{\sqrt{2}} \tilde{\mathcal{A}}_{\sigma,\ell'}(\mathbf{q}_{\parallel})$$

$$= \tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}) + \tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{q}_{\parallel})$$

Represent in Poincaré sphere



$$\frac{1}{\sqrt{2}} \tilde{\mathcal{A}}_{\sigma,\ell}(\mathbf{q}_{\parallel}) + \frac{1}{\sqrt{2}} \tilde{\mathcal{A}}_{\sigma',\ell'}(\mathbf{q}_{\parallel})$$



2. VVB : SOI beyond single beams

Longitudinal component

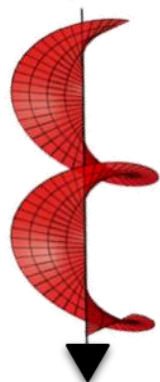
SOI in phase

$$e^{iJ\phi_{q_{\parallel}}}$$



SOI in phase

$$e^{iJ'\phi_{q_{\parallel}}}$$



SOI in phase & **amplitude**

$$\begin{aligned}\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^z(q_{\parallel}) &= \frac{-\sin\theta_q}{2} \left[\tilde{F}_{|\ell|}(q_{\parallel}) e^{iJ\phi_{q_{\parallel}}} + \tilde{F}_{|\ell'|}(q_{\parallel}) e^{iJ'\phi_{q_{\parallel}}} \right] \hat{\mathbf{z}} \\ &= \frac{-\sin\theta_q}{2} e^{i\frac{J'+J}{2}\phi_{q_{\parallel}}} \left[\tilde{F}_{|\ell|}(q_{\parallel}) e^{-i\frac{J'-J}{2}\phi_{q_{\parallel}}} + \tilde{F}_{|\ell'|}(q_{\parallel}) e^{i\frac{J'-J}{2}\phi_{q_{\parallel}}} \right] \hat{\mathbf{z}} \\ &= \frac{-\sin\theta_q}{2} e^{i\frac{\pi}{4}(|\ell'|+|\ell|)} e^{i\left(\frac{J'+J}{2}\right)\phi_{q_{\parallel}}} \left\{ G_{|\ell|,|\ell'|}^+(q_{\parallel}) \cos(\phi_{q_{\parallel}}^{\Delta J}) + G_{|\ell|,|\ell'|}^-(q_{\parallel}) i \sin(\phi_{q_{\parallel}}^{\Delta J}) \right\} \hat{\mathbf{z}}\end{aligned}$$

Definitions

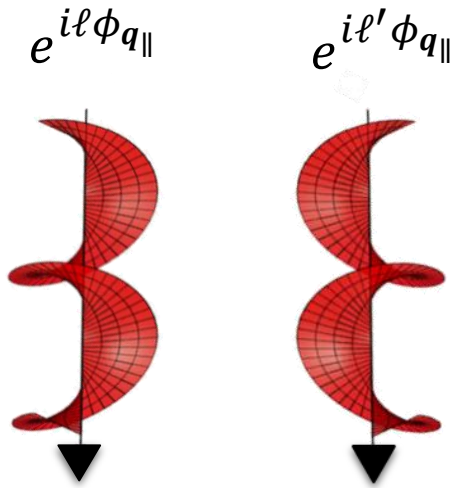
$$\Delta J \equiv J' - J$$

$$\phi_{q_{\parallel}}^{\Delta J} \equiv \frac{\Delta J}{2} \phi_{q_{\parallel}} - \frac{\pi}{2} (|\ell'| - |\ell|)$$

$$G_{|\ell|,|\ell'|}^{\pm}(q_{\parallel}) \equiv F_{|\ell|}(q_{\parallel}) \pm F_{|\ell'|}(q_{\parallel})$$

$$\tilde{F}_{|\ell|}(q_{\parallel}) = (-i)^{|\ell|} F_{|\ell|}(q_{\parallel})$$

2. VVB : SOI beyond single beams



SOI in **polarization**

Transverse component

$$\begin{aligned}\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}) &= \frac{1}{\sqrt{2}} \left[\tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) e^{i\ell\phi_{q_{\parallel}}} \hat{\mathbf{e}}_{\sigma} + \tilde{F}_{|\ell'|}(\mathbf{q}_{\parallel}) e^{i\ell'\phi_{q_{\parallel}}} \hat{\mathbf{e}}_{\sigma'} \right] \\ &= \frac{1}{2} e^{i\frac{\pi}{4}(|\ell'|+|\ell|)} e^{i\left(\frac{\ell'+\ell}{2}\right)\phi_{q_{\parallel}}} \left\{ G_{|\ell|,|\ell'|}^{+}(\mathbf{q}_{\parallel}) \left[\cos(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{\mathbf{x}} + \left(\left(\frac{\sigma'+\sigma}{2} \right) i \cos(\phi_{q_{\parallel}}^{\Delta\ell}) - \frac{\Delta\sigma}{2} \sin(\phi_{q_{\parallel}}^{\Delta\ell}) \right) \hat{\mathbf{y}} \right] \right. \\ &\quad \left. + G_{|\ell|,|\ell'|}^{-}(\mathbf{q}_{\parallel}) i \left[\sin(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{\mathbf{x}} + \left(\left(\frac{\sigma'+\sigma}{2} \right) i \sin(\phi_{q_{\parallel}}^{\Delta\ell}) - \frac{\Delta\sigma}{2} \cos(\phi_{q_{\parallel}}^{\Delta\ell}) \right) \hat{\mathbf{y}} \right] \right\}\end{aligned}$$

To manipulate the polarization, we need $\sigma \neq \sigma'$ and $\ell \neq \ell'$

$$\begin{aligned}\tilde{\mathcal{A}}_{\sigma,\ell,-\sigma,\ell'}^{\parallel}(\mathbf{q}_{\parallel}) &= \frac{1}{2} e^{i\frac{\pi}{4}(|\ell'|+|\ell|)} e^{i\left(\frac{\ell'+\ell}{2}\right)\phi_{q_{\parallel}}} \left\{ G_{|\ell|,|\ell'|}^{+}(\mathbf{q}_{\parallel}) \left[\cos(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{\mathbf{x}} - \frac{\Delta\sigma}{2} \sin(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{\mathbf{y}} \right] \right. \\ &\quad \left. + G_{|\ell|,|\ell'|}^{-}(\mathbf{q}_{\parallel}) i \left[\sin(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{\mathbf{x}} - \frac{\Delta\sigma}{2} \cos(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{\mathbf{y}} \right] \right\}\end{aligned}$$

Definitions

$$\Delta\sigma \equiv \sigma' - \sigma$$

$$\Delta\ell \equiv \ell' - \ell$$

$$\phi_{q_{\parallel}}^{\Delta\ell} \equiv \frac{\Delta\ell}{2} \phi_{q_{\parallel}} - \frac{\pi}{2} (|\ell'| - |\ell|)$$

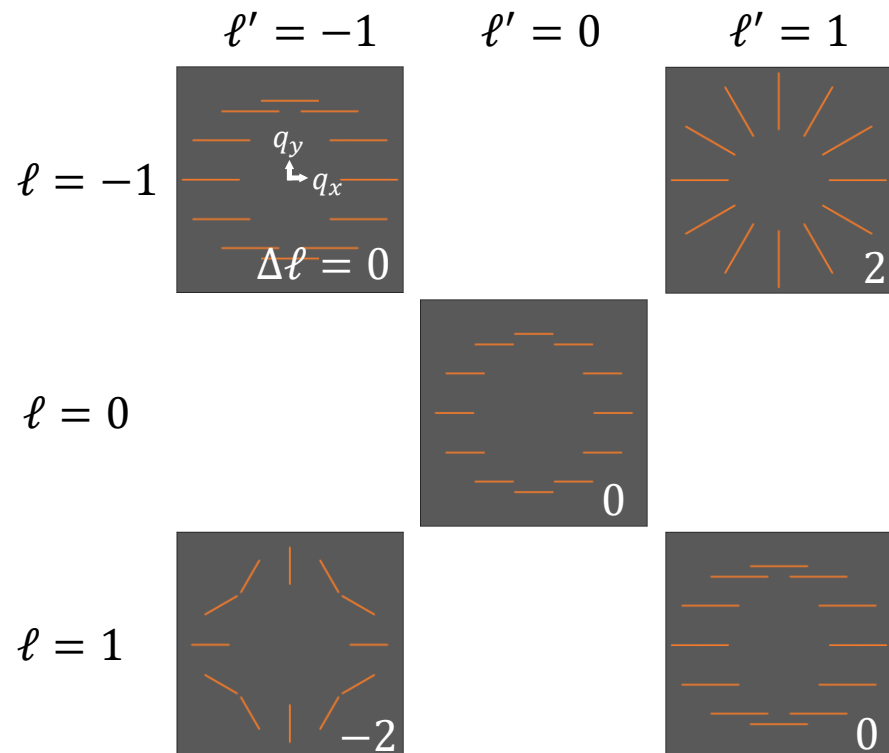
2. VVB : even number of AM difference - symmetry

$$\Delta\sigma = -2 \begin{cases} \mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}) \propto F_{|\ell|}(q_{\parallel}) \left\{ \cos(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{x} - \frac{\Delta\sigma}{2} \sin(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{y} \right\} \\ \mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}) \propto F_{|\ell|}(q_{\parallel}) \cos(\phi_{q_{\parallel}}^{\Delta J}) \hat{z} \end{cases}$$

$$\phi_{q_{\parallel}}^{\Delta J} \equiv \frac{\Delta J}{2} \phi_{q_{\parallel}} - \frac{\pi}{2} (|\ell'| - |\ell|)$$

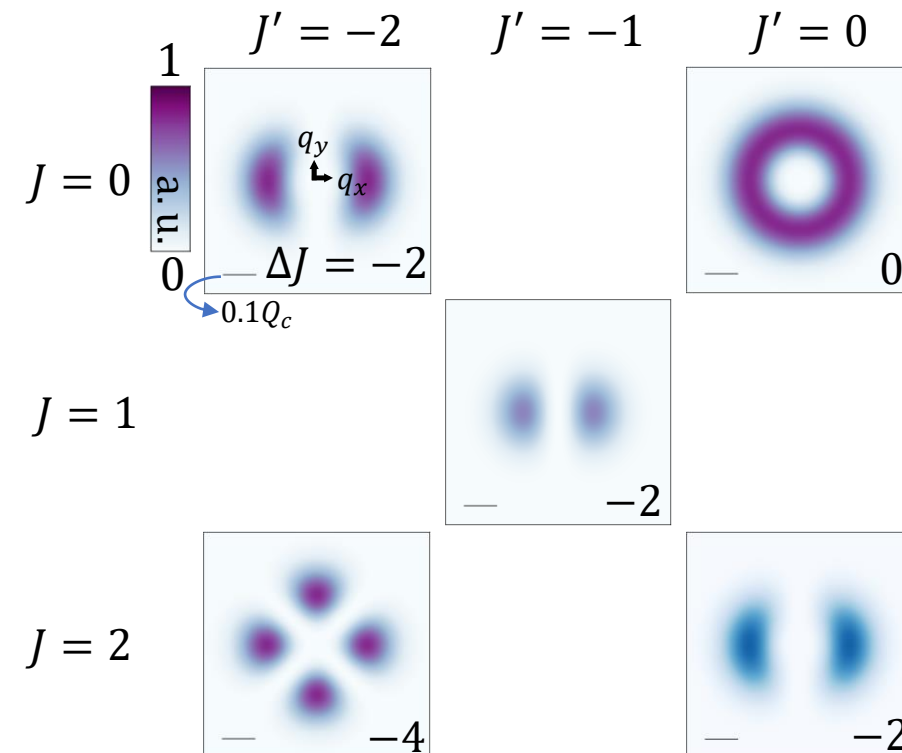
$$\phi_{q_{\parallel}}^{\Delta\ell} \equiv \frac{\Delta\ell}{2} \phi_{q_{\parallel}} - \frac{\pi}{2} (|\ell'| - |\ell|)$$

Polarization trajectory



Symmetry when $\Delta\ell$ is an **even** number

Longitudinal intensity



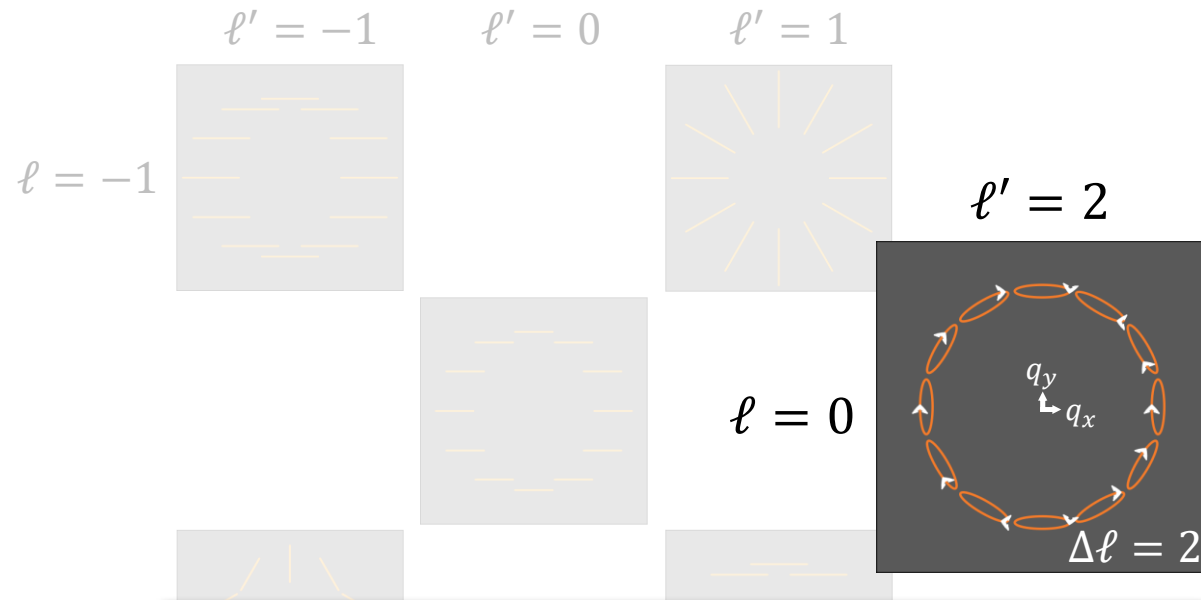
Symmetry when ΔJ is an **even** number

2. VVB : even number of AM difference - symmetry ($|\ell| \neq |\ell'|\rangle$)

$$\Delta\sigma = -2 \begin{cases} \mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}) \propto \frac{1}{2} \left\{ G_{|\ell|,|\ell'|}^{+}(\mathbf{q}_{\parallel}) \left[\cos(\phi_{\mathbf{q}_{\parallel}}^{\Delta\ell}) \hat{\mathbf{x}} - \frac{\Delta\sigma}{2} \sin(\phi_{\mathbf{q}_{\parallel}}^{\Delta\ell}) \hat{\mathbf{y}} \right] + G_{|\ell|,|\ell'|}^{-}(\mathbf{q}_{\parallel}) i \left[\sin(\phi_{\mathbf{q}_{\parallel}}^{\Delta\ell}) \hat{\mathbf{x}} - \frac{\Delta\sigma}{2} \cos(\phi_{\mathbf{q}_{\parallel}}^{\Delta\ell}) \hat{\mathbf{y}} \right] \right\} \\ \mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}) \propto \frac{1}{2} \left\{ G_{|\ell|,|\ell'|}^{+}(\mathbf{q}_{\parallel}) \cos(\phi_{\mathbf{q}_{\parallel}}^{\Delta J}) + G_{|\ell|,|\ell'|}^{-}(\mathbf{q}_{\parallel}) i \sin(\phi_{\mathbf{q}_{\parallel}}^{\Delta J}) \right\} \hat{\mathbf{z}} \end{cases}$$

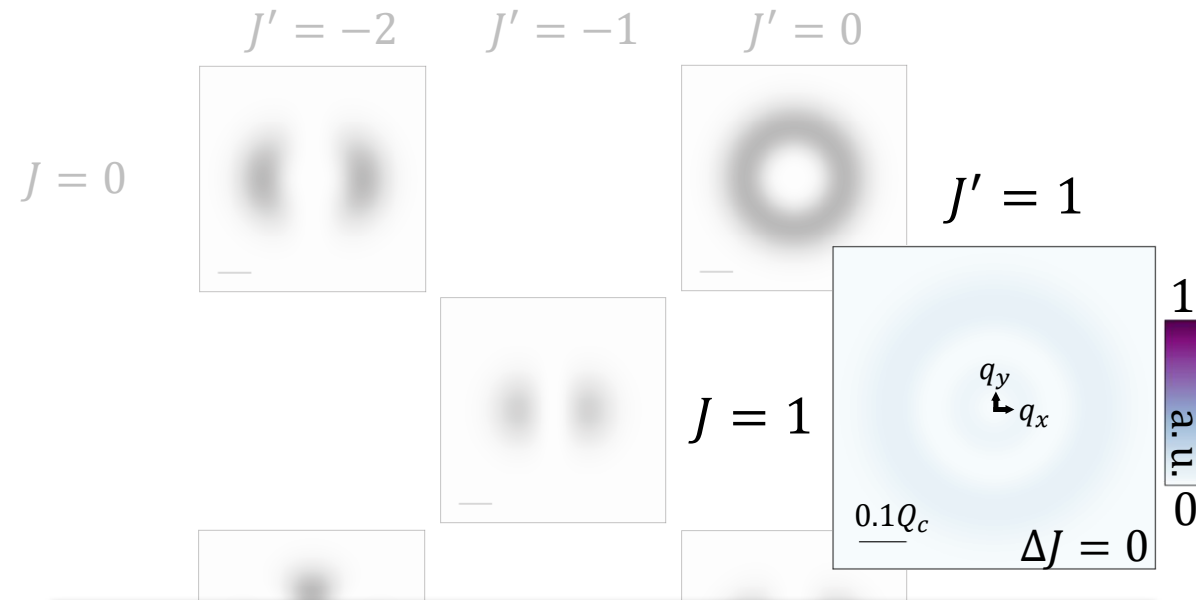
$$G_{|\ell|,|\ell'|}^{\pm}(\mathbf{q}_{\parallel}) \equiv F_{|\ell|}(\mathbf{q}_{\parallel}) \pm F_{|\ell'|}(\mathbf{q}_{\parallel})$$

Polarization trajectory



1. Symmetry since $\Delta\ell$ is an **even** number
2. Elliptical since $G_{|\ell|,|\ell'|}^{+}(\mathbf{q}_{\parallel}) \gg G_{|\ell|,|\ell'|}^{-} \neq 0$

Longitudinal intensity

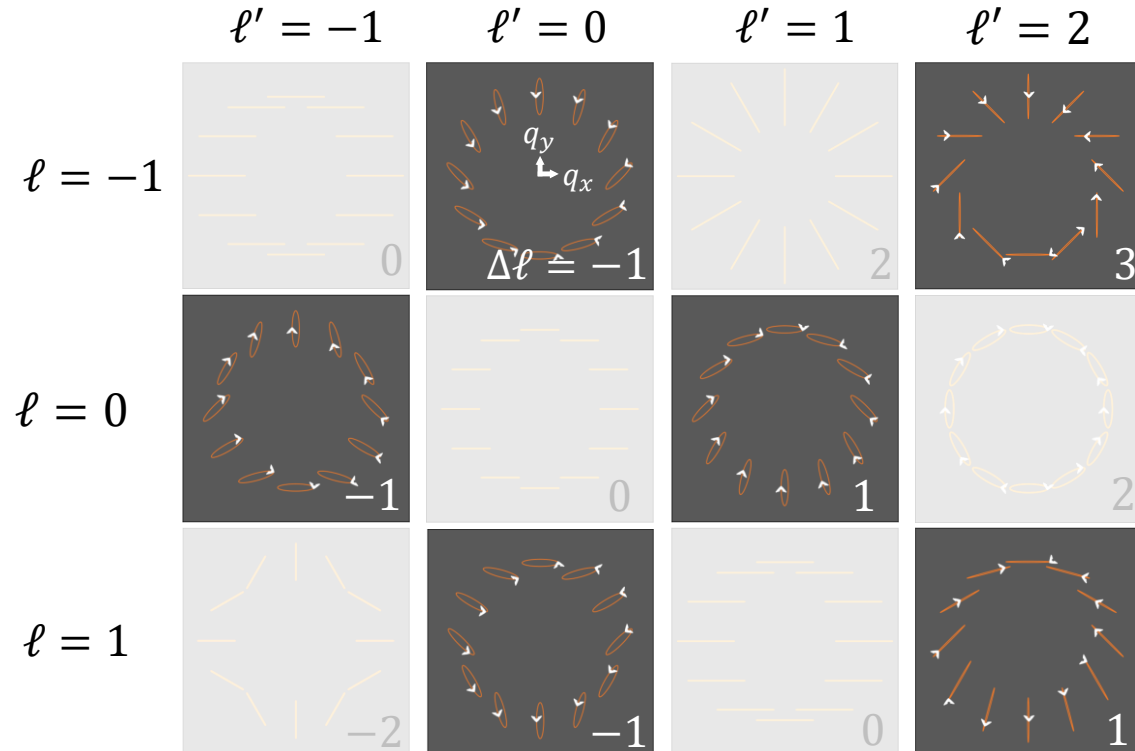


1. Symmetry since ΔJ is an **even** number
2. Additional pattern since $G_{|\ell|,|\ell'|}^{+}(\mathbf{q}_{\parallel}) \gg G_{|\ell|,|\ell'|}^{-} \neq 0$

2. VVB : odd number of AM difference - asymmetry

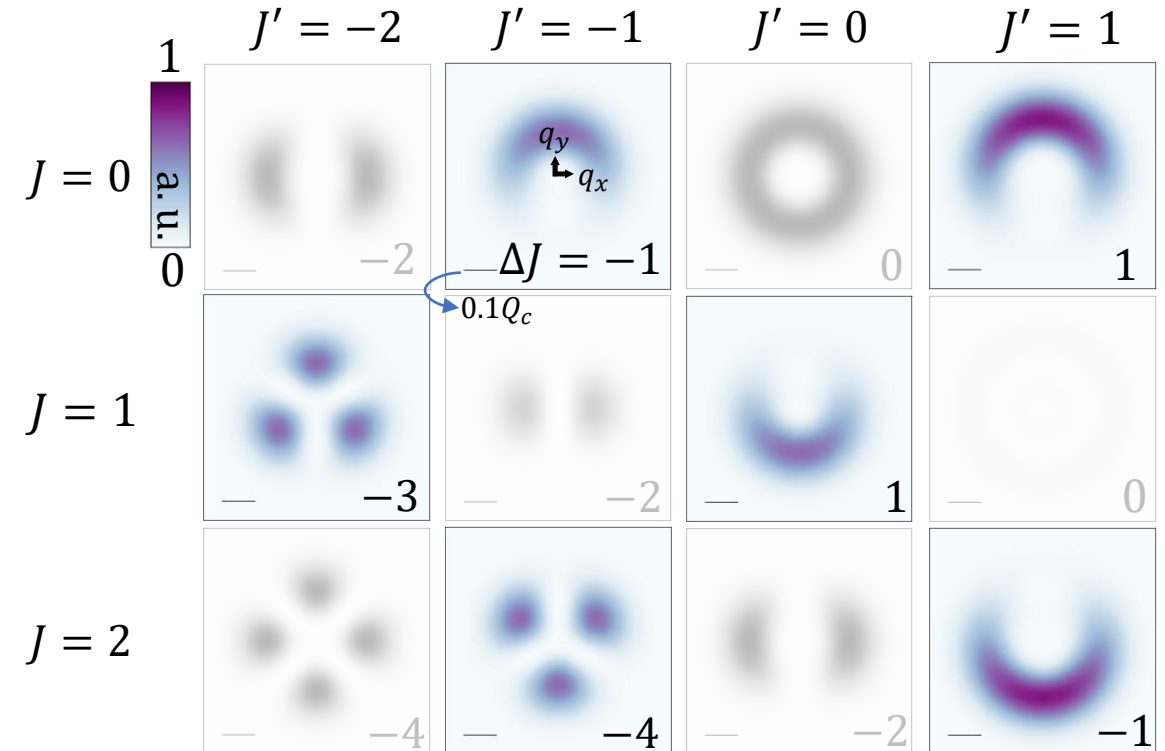
$$\Delta\sigma = -2 \begin{cases} \mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}) \propto \frac{1}{2} \left\{ G_{|\ell|,|\ell'|}^{+}(q_{\parallel}) \left[\cos(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{\mathbf{x}} - \frac{\Delta\sigma}{2} \sin(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{\mathbf{y}} \right] + G_{|\ell|,|\ell'|}^{-}(q_{\parallel}) i \left[\sin(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{\mathbf{x}} - \frac{\Delta\sigma}{2} \cos(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{\mathbf{y}} \right] \right\} \\ \mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}) \propto \frac{1}{2} \left\{ G_{|\ell|,|\ell'|}^{+}(q_{\parallel}) \cos(\phi_{q_{\parallel}}^{\Delta J}) + G_{|\ell|,|\ell'|}^{-}(q_{\parallel}) i \sin(\phi_{q_{\parallel}}^{\Delta J}) \right\} \hat{\mathbf{z}} \end{cases}$$

Polarization trajectory



Asymmetry when $\Delta\ell$ is an **odd** number

Longitudinal intensity



Asymmetry when ΔJ is an **odd** number

1. Introduction
2. Vector vortex beam (VVB) : SOI beyond single beams
3. Optical transitions
 - Directional momentum-selective excitation
 - Selective excitation

Summary and outlook

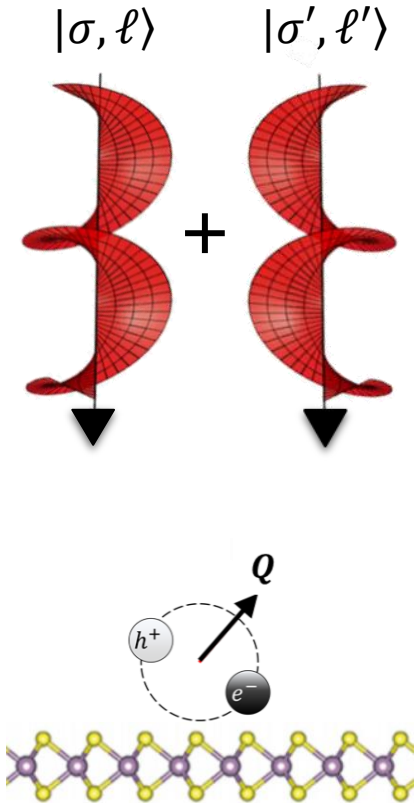
3. Directional momentum-selective excitation

The transition rate is obtained as

$$\Gamma_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}) = \frac{2\pi}{\hbar} \left| \tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}) \right|^2 \rho(\hbar\omega = E_{S,Q}^X), \quad S = B+, B-, G$$

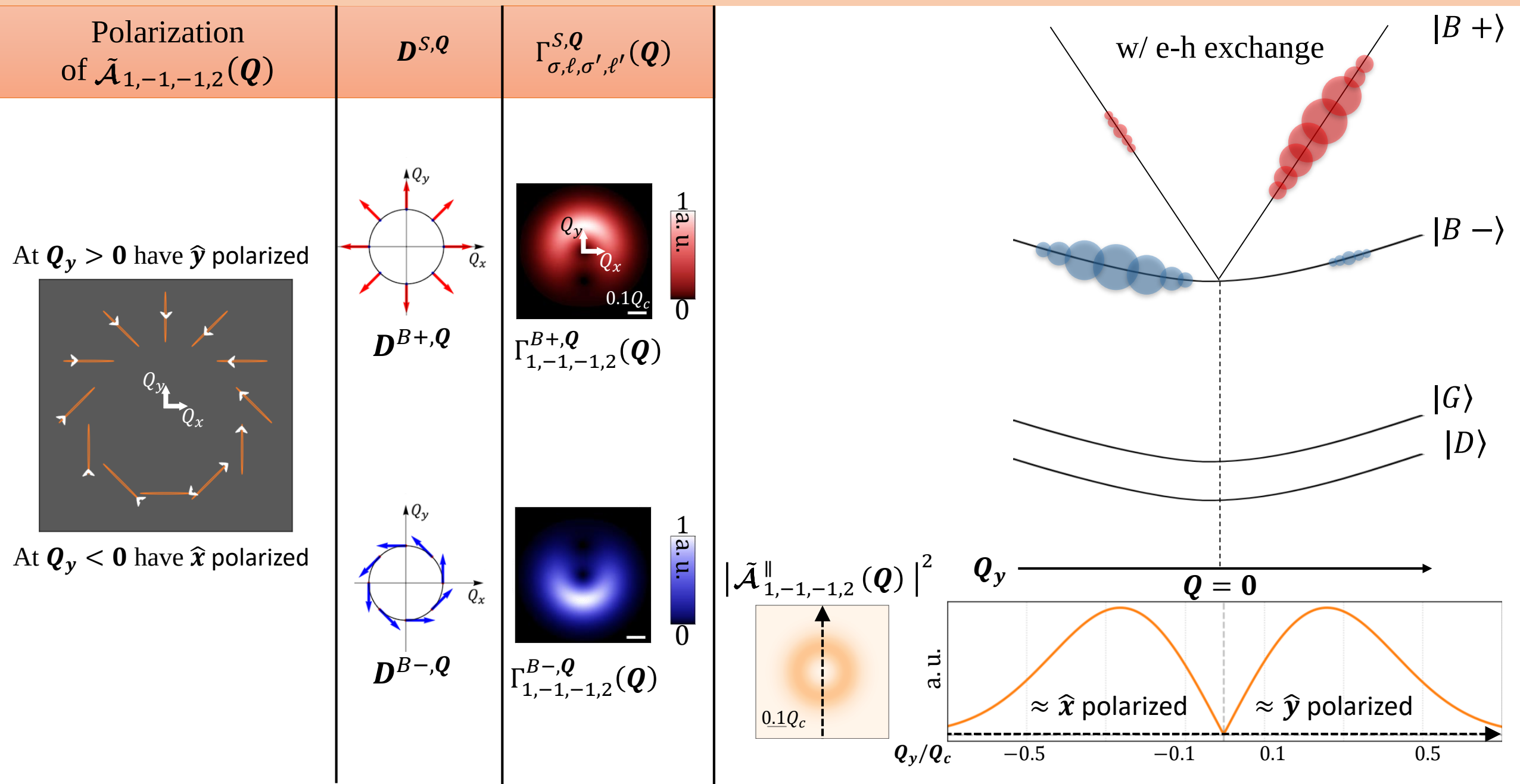
The optical matrix element is

$$\tilde{M}_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}) = \frac{E_g}{2i\hbar} (\tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}(\mathbf{Q}) \cdot \mathbf{D}^{S,Q*})$$

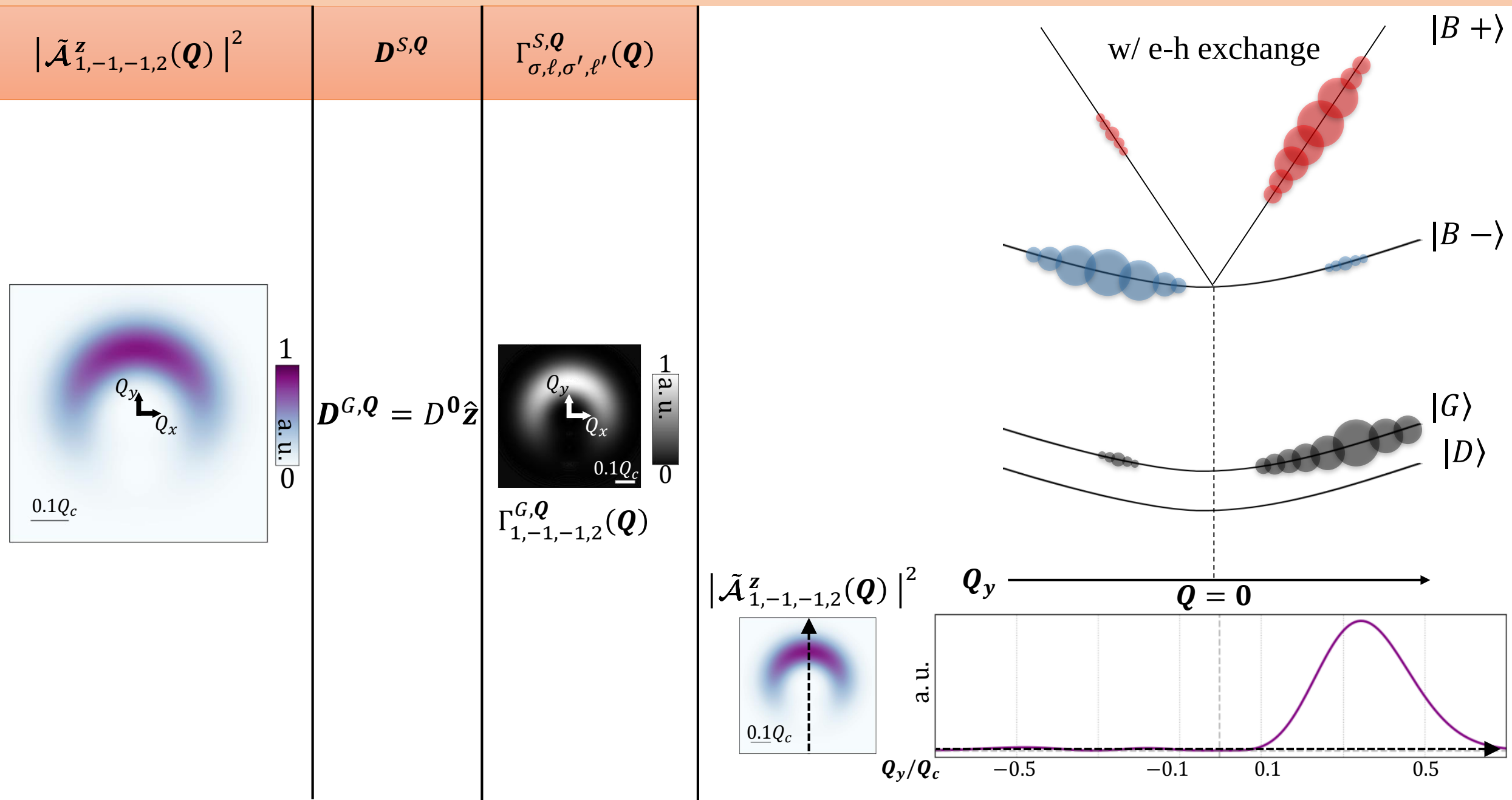


$\mathbf{D}^{S,Q}$: transition dipole moment

3. Directional momentum-selective excitation (BXs)



3. Directional momentum-selective excitation (GXs)



4. Effect of relative phase α

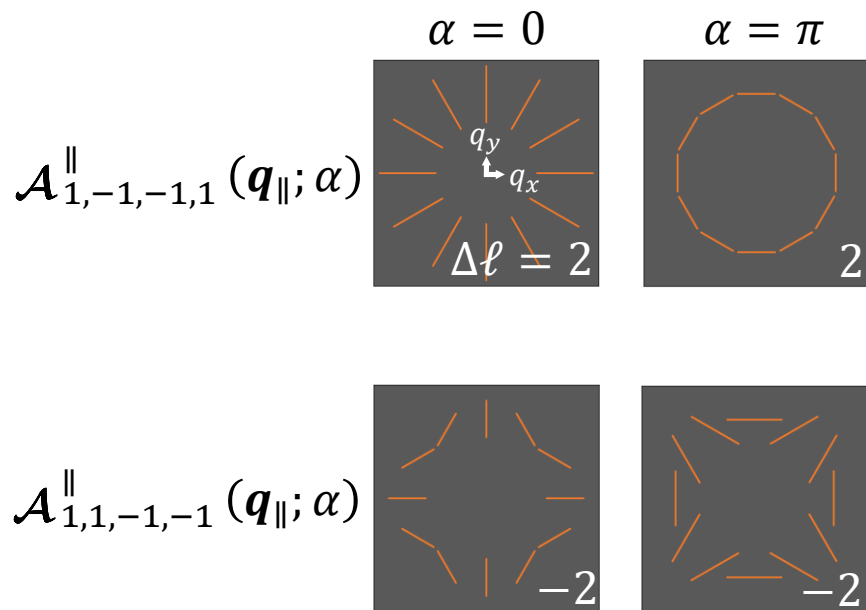
VVB :
$$\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = \frac{1}{\sqrt{2}} \tilde{\mathcal{A}}_{\sigma,\ell}(\mathbf{q}_{\parallel}) + e^{i\alpha} \frac{1}{\sqrt{2}} \tilde{\mathcal{A}}_{\sigma',\ell'}(\mathbf{q}_{\parallel}) = \tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha) + \tilde{\mathcal{A}}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{q}_{\parallel}; \alpha)$$

$$\Delta\sigma = -2 \begin{cases} \mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha) \propto \frac{1}{2} \left\{ G_{|\ell|,|\ell'|}^{+}(\mathbf{q}_{\parallel}) \left[\cos(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{x} - \frac{\Delta\sigma}{2} \sin(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{y} \right] + G_{|\ell|,|\ell'|}^{-}(\mathbf{q}_{\parallel}) i \left[\sin(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{x} - \frac{\Delta\sigma}{2} \cos(\phi_{q_{\parallel}}^{\Delta\ell}) \hat{y} \right] \right\} \\ \mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha) \propto \frac{1}{2} \left\{ G_{|\ell|,|\ell'|}^{+}(\mathbf{q}_{\parallel}) \cos(\phi_{q_{\parallel}}^{\Delta J}) + G_{|\ell|,|\ell'|}^{-}(\mathbf{q}_{\parallel}) i \sin(\phi_{q_{\parallel}}^{\Delta J}) \right\} \hat{z} \end{cases}$$

$$\phi_{q_{\parallel}}^{\Delta J} \equiv \frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha}{2} - \frac{\pi}{2} (|\ell'| - |\ell|)$$

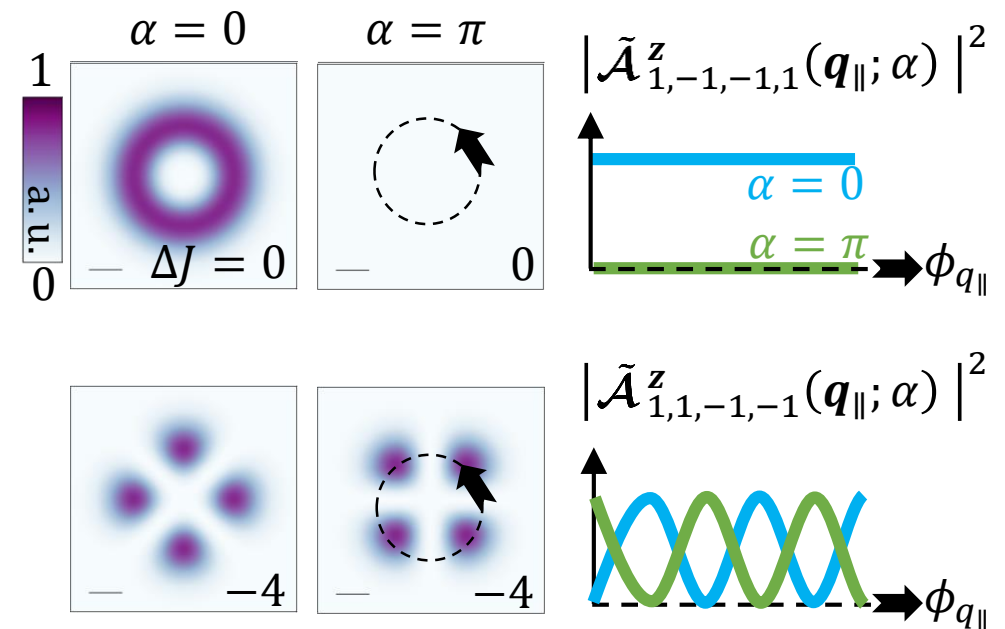
$$\phi_{q_{\parallel}}^{\Delta\ell} \equiv \frac{\Delta\ell}{2} \phi_{q_{\parallel}} + \frac{\alpha}{2} - \frac{\pi}{2} (|\ell'| - |\ell|)$$

Polarization trajectory



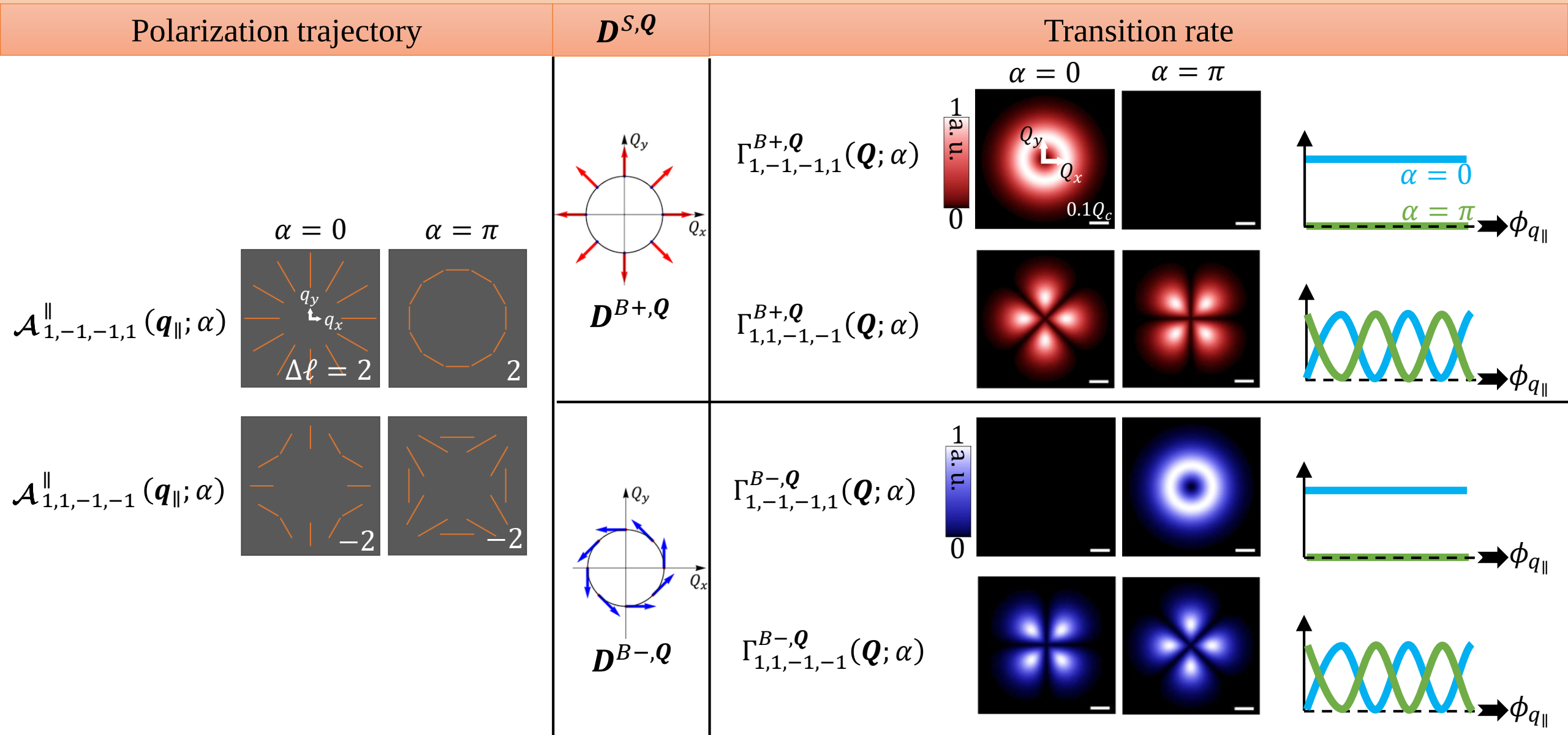
α rotate the orientation

Longitudinal intensity



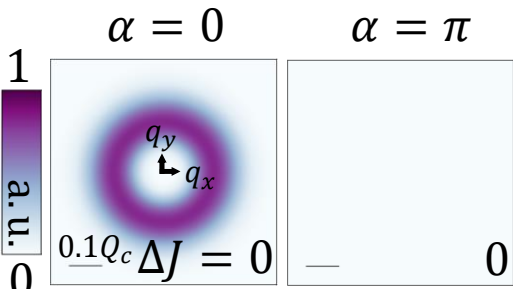
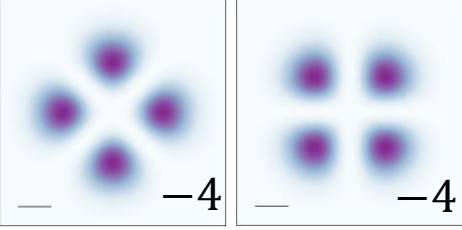
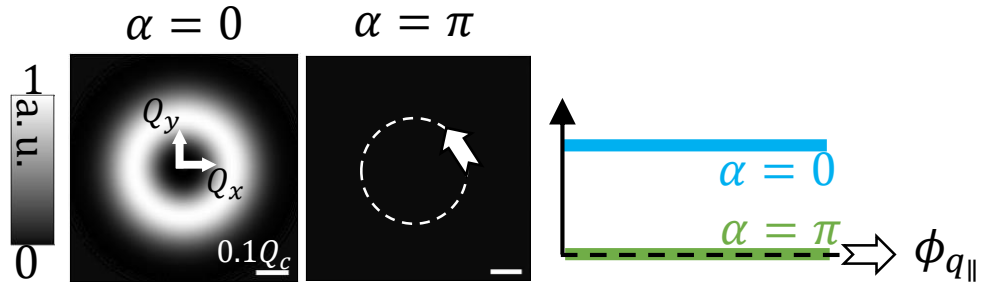
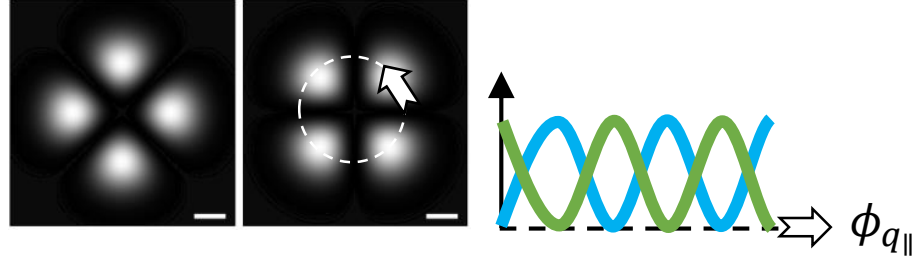
Only when $\Delta J = 0$ can α change the overall value

4. Selective excitation with α



Only $\Delta\ell = 2$ can α change the excitation

4. Selective excitation with α

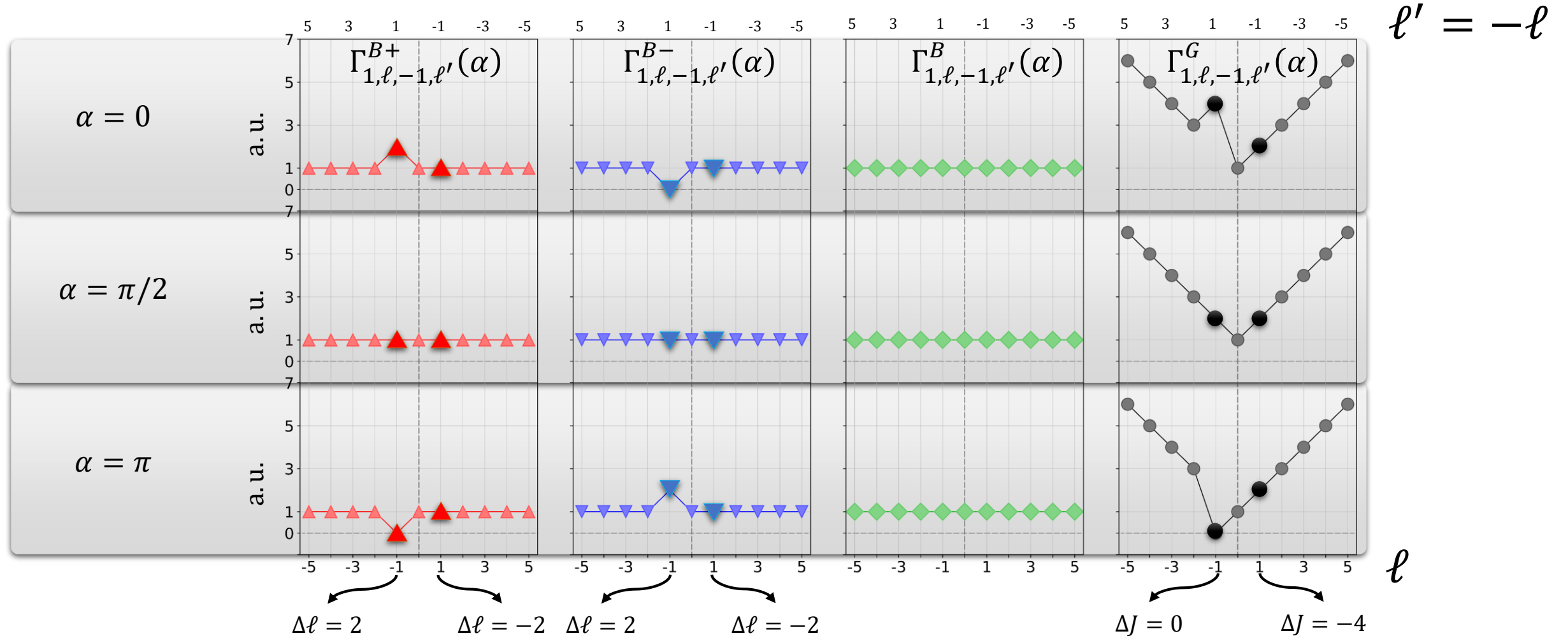
Longitudinal intensity	$D^{S,Q}$	Transition rate
$\tilde{\mathcal{A}}_{1,-1,-1,1}^z(\mathbf{q}_{\parallel}; \alpha) ^2$  $\tilde{\mathcal{A}}_{1,1,-1,-1}^z(\mathbf{q}_{\parallel}; \alpha) ^2$ 	$D^{G,Q}$ $=$ $D^0 \hat{\mathbf{z}}$	$\Gamma_{1,-1,-1,1}^{G,Q}(\mathbf{Q}; \alpha)$  $\Gamma_{1,1,-1,-1}^{G,Q}(\mathbf{Q}; \alpha)$ 

Only $\Delta J = 0$ can α change the excitation

4. Selective excitation with α

Total transition rate : $\Gamma_{\sigma,\ell,\sigma',\ell'}^S(\alpha) = \sum_Q \Gamma_{\sigma,\ell,\sigma',\ell'}^{S,Q}(\mathbf{Q}; \alpha)$

Total transition rate for total BXs : $\Gamma_{\sigma,\ell,\sigma',\ell'}^B(\alpha) = \sum_{B\pm} \Gamma_{\sigma,\ell,\sigma',\ell'}^{B\pm}(\alpha)$



Summary and outlook

Summary :

- The **TAM difference and OAM difference** independently determine the angular geometry of the **longitudinal and transverse** components, manifested as **intensity and polarization** distributions, respectively. This governing rule remains valid beyond the equal-OAM condition ($|\ell| = |\ell'|$) provided that the constituent beams maintain sufficient spatial overlap.
- Matching between the optical angular geometry and the orientation of exciton transition dipole moments enables **directional momentum-selective excitation** of excitons in center-of-mass momentum space.
- The relative phase α generally rotates the optical angular geometry and the corresponding transition-rate distribution. Under the special SOI condition, $\Delta\ell = 2$ and $\Delta J = 0$, however, relative phase α acts as an **on/off switch for exciton excitation**.

Outlook

- Selective excitation of excitons : Observe the evolution of exciton populations generated by tailored vector vortex beams through time-resolved photoluminescence imaging.
- Exciton wave packets : Vector vortex beams actually carry average angular momentum in their phase, which can be further utilized to engineer exciton wave packets.

Acknowledgment

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Thank you for listening